ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 1 · PRACTICE C · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Electronics and Digital Concepts for Robotics paper 1 · practice c.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and neat, recognisable diagrams.

SECTION A

QUESTION 1

1 1.1 B ✓ 1.2 D ✓ 1.3 A ✓ 1.4 C ✓ 1.5 A ✓ 1.6 B ✓ 1.7 D ✓ 1.8 C ✓ 1.9 A ✓ 1.10 B ✓ (10)

[10]

QUESTION 2

2 2.1 True ✓ 2.2 False ✓ 2.3 False ✓ 2.4 True ✓ 2.5 True ✓ 2.6 False ✓ 2.7 True ✓ 2.8 True ✓ 2.9 False ✓ 2.10 True ✓ (10)

[10]

QUESTION 3

3 3.1 F ✓ 3.2 C ✓ 3.3 H ✓ 3.4 G ✓ 3.5 E ✓ 3.6 J ✓ 3.7 B ✓ 3.8 A ✓ 3.9 D ✓ 3.10 I ✓ (10)

[10]

TOTAL SECTION A: [30]

SECTION B

QUESTION 4

4.1 $4{,}7 \times 10^{-4}$ ✓ (1)

4.2 $38\,000$ ✓ (1)

4.3 $v = d \div t$ ✓; $t = 20 \div 60 = 0{,}33$ h ✓✓; $v = 2{,}4 \div (20 \div 60)$ ✓ $= 7{,}2$ km/h ✓ (5)

4.4 $W = m \times g = 15 \times 9{,}8$ ✓ $= 147$ N ✓ (2)

4.5 $\theta = 60 \times \pi \div 180$ ✓ $= 1{,}05$ rad ✓; $s = r \times \theta$ ✓ $= 0{,}5 \times 1{,}047$ ✓ $= 0{,}52$ m ✓ (5)

4.6 $V = I \times R = 0{,}25 \times 48$ ✓ $= 12$ V ✓ (2)

4.7 Any FOUR ✓✓✓✓: electrical; chemical; mechanical (kinetic / potential); heat (thermal); light; sound; nuclear. (4)

[20]

QUESTION 5

5.1 5.1.1 Diode ✓ 5.1.2 Light-emitting diode (LED) ✓ 5.1.3 Fuse ✓ 5.1.4 Switch (single-pole, single-throw / push-to-make) ✓ 5.1.5 Lamp ✓ (5)

5.2 Circle (optional) with the base as a straight line ✓, collector ✓ and emitter ✓ lines; arrow on the emitter pointing outwards ✓; terminals labelled B, C, E ✓ (5)

[10]

QUESTION 6

6.1 Direct current flows in one direction only (e.g. from a battery) ✓; alternating current changes direction regularly, many times a second (e.g. the 50 Hz mains supply) ✓. (2)

6.2 Yellow ✓ violet ✓ red ✓ gold ✓ (4)

6.3 6.3.1 Light sensor, e.g. night light / line sensor ✓ 6.3.2 Volume or speed control, setting a level ✓ 6.3.3 Protects a circuit against too much current ✓ 6.3.4 Smoothing / filtering, storing charge, timing ✓ (4)

[10]

QUESTION 7

7.1 $13 = 8 + 4 + 1 \text{ (or repeated division by 2)} \checkmark = 1101_2 \checkmark$ (2)

7.2.1 Correct NAND symbol (AND shape with a small circle at the output) ✓; truth table ✓✓✓✓ (00 → 1, 01 → 1, 10 → 1, 11 → 0) (5)

7.2.2 Correct NOR symbol (OR shape with a small circle at the output) ✓; truth table ✓✓✓✓ (00 → 1, 01 → 0, 10 → 0, 11 → 0) (5)

7.3 An electronic circuit ✓ with one or more inputs and one output ✓ that follows a logical rule (AND, OR, NOT …), using two states, 0 and 1 ✓. (3)

[15]

QUESTION 8

8.1 Any THREE ✓✓✓: safety glasses; antistatic wrist strap; closed shoes; overall / lab coat; heat-resistant gloves when soldering; fume extractor / mask. (3)

8.2 Any THREE ✓✓✓: wear an earthed antistatic wrist strap; work on an antistatic mat; keep components in antistatic bags until needed; touch an earthed object before handling components; avoid synthetic clothing and carpets; hold components by their edges. (3)

8.3 Any FIVE ✓✓✓✓✓: select the correct function and range before connecting; start on the highest range if the value is unknown; never measure resistance on a live circuit; connect an ammeter in series and a voltmeter in parallel; check the leads for damage; keep fingers behind the probe guards; do not exceed the meter's rating; switch off after use. (5)

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8.4 A hazard is anything that can cause harm ✓✓ (e.g. a hot soldering iron). A risk is the chance (likelihood) that the hazard will actually cause harm, and how serious it would be ✓✓. (4)

[15]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE